Applied Economics

Instrumental Variables

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Plan for this Lecture

1. Simple Case: Bivariate Regression with one instrument
2. Example in R
3. General Case: Multivariate Regression with one or more endogenous regressors
4. Example in R

Motivation

We discussed different sources of endogeneity, i.e. a violation of MLR.4: $E[u_i|\boldsymbol{X}_i] \neq 0$

Instrumental Variable (IV) estimation is an approach to get a consistent estimator if u_i and $\boldsymbol{X}_i$ are correlated.

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Terminology: we call a variable endogenous [exogenous] if it is correlated [not correlated] with the error term.

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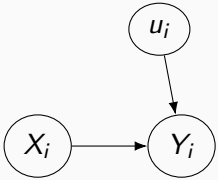
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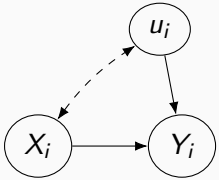
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Idea: Find a variable (the instrument) that is **correlated with the endogenous regressor** (say, X_{1i}), **but uncorrelated with** u_i
$\rightarrow$ decompose the variation in X_{1i} into an endogenous and an exogenous part
$\rightarrow$ extract the part that is not correlated with u_i, and exploit only that part in the estimation of $\boldsymbol{\beta}$

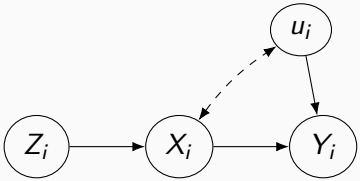
Illustration: Instrumental Variables (IV) approach



X_i is exogenous



X_i is endogenous



Idea IV Estimation

Regression with one Endogenous Regressor and one Instrument

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } E[X_i u_i] \neq 0$$

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1. **Relevance**: $cov(Z_i, X_i) \neq 0$
   The IV is correlated with the endogenous explanatory variable
   $\rightarrow$ can be used to extract variation in X_i
2. **Exogeneity**: $cov(Z_i, u_i) = 0$
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Which of the two assumptions can be empirically tested?

The IV assumptions

Testing for instrument **relevance**:

First Stage Regression Model: $X_i = \pi_0 + \pi_1 Z_i + v_i$

$H_0 : \pi_1 = 0 \rightarrow$ test of relevance condition $(\pi_1 = \frac{cov(X_i, Z_i)}{Var(Z_i)})$

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Exogeneity of an instrument is *an untestable assumption*

- needs to be argued for using theory (like MLR.4)
- involves arguing that the IV cannot be related to anything in u_i which is often tricky.

$$\begin{aligned} Cov(Z_i, Y_i) &= Cov(Z_i, \beta_0 + \beta_1 X_i + u_i) \\ &= \beta_1 Cov(Z_i, X_i) + Cov(Z_i, u_i) \end{aligned}$$

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This identifies the population parameter β_1.

IV Identification and Estimation

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This identifies the population parameter β_1.

Using the sample analogue yields the IV Estimator:

$$
\hat{\beta}_1^{IV} = \frac{\widehat{Cov}(Z_i, Y_i)}{\widehat{Cov}(Z_i, X_i)}
$$

Two-Stage Least Squares (2SLS)

In the case with a single endogenous regressor and a single instrument, the IV estimator is **equivalent** to the 2SLS estimator.

This estimator illustrates the intuition that IV *extracts* the exogenous variation in the endogenous regressor.

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First Stage regression: $X_i = \overbrace{\pi_0 + \pi_1 Z_i}^{\text{uncorrelated with } u_i} + v_i$

$\rightarrow$ decomposes variance in X_i into exogenous ($\pi_0 + \pi_1 Z_i$) and endogenous part (v_i).

$\rightarrow$ compute fitted values $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i$.

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Second Stage regression: $Yi = \beta_0 + \beta_1 \hat{X}_i + u_i$

$\rightarrow$ Replace X_i with fitted values from the first stage in the regression model of interest.

$\rightarrow$ OLS estimation of the second stage model yields 2SLS estimator $\hat{\beta}_0^{2SLS}$, $\hat{\beta}_1^{2SLS}$

2SLS estimator: Properties

Under the assumptions of

- linearity in parameters (MLR.1),
- an i.i.d. random sample (MLR.2),
- instrument relevance (instead of MLR.3) and
- instrument exogeneity (instead of MLR.4),

the 2SLS estimator is consistent and asymptotically normal.

Example in R

Education and wage again

«include = F»= library(wooldridge) library(sandwich) library(lmtest) library(car)

data("mroz") data on education and wages of married women in the US mroz <- mroz[!is.na(mroz$lwage),]

@

Estimate simple regression model by **OLS** «»=

$ols_est <- lm(lwage \ educ, data = mroz) coeftest(ols_est, vcov. = vcovHC(ols_est, "HC1"))$@

One additional year of education on average associated with increase in wage by 10.9%

Using maternal education as an IV

Is mother's education a suitable instrument?

Using maternal education as an IV

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First Stage Regression (Relevance) «»=

$first_st < -lm(educ\ motheduc, data = mroz)coeftest(first_st, vcov. = vcovHC(first_st, "HC1"))$@

Using maternal education as an IV

What about exogeneity?

Using maternal education as an IV

Second Stage Regression *(manually)* «»= mroz$fitted_educ < -first_stfitted.values$ get fitted values from first stage $second_st < -lm(lwage\ fitted_educ, data = mroz)coeftest(second_st, vcov. = vcovHC(second_st, "HC1"))$

@ One additional year of education on average increases wage by 3.9% → less than OLS estimate (which is likely biased upward).

When running the two stages separately, standard errors in the 2. stage are incorrect!

Using maternal education as an IV

Using built-in IV estimator gives correct standard errors

```
> install.packages("estimatr")
> library(estimatr)
> tsls <- iv_robust(lwage ~ educ | motheduc, data = mroz, se_type = "HC1",
diagnostics = T)
> summary(tsls)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high	df
(Intercept)	0.702	0.494	1.422	0.156	-0.268	1.673	426
educ	0.039	0.039	0.982	0.327	-0.039	0.116	426

Practice

Run the First Stage Regression and get the IV estimate with `iv_robust()` with *fatheduc* as instrument.

How do the results compare to the case with *motheduc* as IV?

Variance of the IV Estimator

Let's look at the variance of $\hat{\beta}_1^{IV}$ under conditional *homoskedasticity* $E[u_i^2|Z_i] = \sigma_u^2$ (unrealistic in practice, but helpful to study drivers of the variance.)

$$Var(\hat{\beta}_1^{IV}) = \frac{\sigma_u^2}{n\sigma_X^2\rho_{x,z}^2}$$

It increases in error variance σ_u^2, decreases in sample size n and in variance of the regressor σ_X^2. Apart from $\rho_{x,z}^2$ this is identical to the OLS variance.

$\rho_{x,z}$ is the population correlation between X and Z.

Since $\rho_{X,Z} < 1$, the **IV variance is larger than the OLS variance** $Var(\hat{\beta}_1^{OLS})$.

- because we only use a part of the total variation in X_i in the IV estimation
- the stronger the correlation between X and Z, the smaller is the increase in the variance (more variation is extracted by Z_i and used in the estimation).

Instrument Relevance and Weak Instruments

If the correlation betw. Z_i and X_i is small, we have a weak instrument problem, that yields very imprecise IV estimates

Rule of thumb for sufficiently strong instrument: F-statistic in the first stage > 10.

New research suggests that the threshold value for the F-test should be chosen much larger in practice, ≈ 104.

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Weak instruments are also an issue for consistency if exogeneity is not fully satisfied:

$$plim(\hat{\beta}_1^{OLS}) = \beta_1 + Cov(X, u)/Var(X)$$
$$plim(\hat{\beta}_1^{IV}) = \beta_1 + Cov(Z, u)/Cov(Z, X)$$

IV estimator is consistent if exogeneity is satisfied: $Cov(Z, u) = 0$

If exogeneity does not hold, inconsistency of IV is stronger the weaker the correlation of Z and X!

OLS vs. IV

If MLR.4 is violated, we can choose between inconsistent, but precise OLS estimator and a consistent, but less precise IV estimator.
$\rightarrow$ **Bias–Variance-Tradeoff**

Best choice depends on the situation.

- Do IV if you think endogeneity is sufficiently important.
  Otherwise, inconsistency of OLS is small, and it might still be close to the true value; whereas IV is correct in infinite samples, but might be pretty far off the true value in finite samples.
- Try to find **strong** instruments, e.g. in the bivariate case an IV where $|corr(X_i, Z_i)|$ is large.

General Case: Multiple regression model with one or more endogenous regressors

General Case

General linear model with endogeneity: $Y_i = \boldsymbol{X}_i'\boldsymbol{\beta} + ui, \quad \text{with } E[\boldsymbol{X}_i u_i] \neq 0$

We partition the elements of $\boldsymbol{X}_i$ (k x 1) into

- endogenous regressors $\boldsymbol{X}_{1i}$ (k_1 x 1 - vector) which we try to instrument and
- exogenous regressors $\boldsymbol{X}_{2i}$ (k_2 x 1 - vector)

The exogenous regressors serve as 'instruments for themselves':

included instruments $\boldsymbol{Z}_{2i} = \boldsymbol{X}_{2i}$ (k_2 x 1 - vector)

For the endogenous regressors, we need variables from outside the model as IVs:

excluded instruments $\boldsymbol{Z}_{1i}$ (l_1 x 1 - vector)

We write the (l x 1)-vector of instrumental variables: $\boldsymbol{Z}_i = \begin{pmatrix} \boldsymbol{Z}_{1i} \\ \boldsymbol{Z}_{2i} \end{pmatrix}$

We need at least as many excluded instruments as endogenous regressors: $l_1 \geq k_1$

If $l_1 = k_1$ (as many IVs as endogenous regressors), the model is called *just-identified*.

If $l_1 > k_1$ (more IVs than endogenous regressors), the model is called *over-identified*.

Conditions for suitable IVs

1. Exogeneity: uncorrelated with error u_i
2. Relevance: partially correlated with endogenous explanatory variable

First Stage and Relevance

First stage regression(s): For each endogenous regressor, run a first stage regression of that regressor on *all included and excluded instruments*:

$$X_{ji} = \pi_0 + \underbrace{\pi_1 Z_{1i} + ... + \pi_{l_1} Z_{l_1 i}}_{\text{excluded IVs}} + \underbrace{\boldsymbol{X}'_{2i}\gamma}_{\substack{\text{exogenous regressors/} \\ \text{included IVs}}} + v_i, \quad \text{for } j = 1, ..., k_1.$$

Testing Relevance:

$H_0 : \pi_1 = \pi_2 = ... = \pi_{L_1} = 0$:

$\rightarrow$ F-Test of joint significance of the excluded IVs for the endog. regressor.

Estimate the first stage(s) and obtain the fitted values of the endogenous regressor(s): $(\hat{X}_{1i}, ..., \hat{X}_{ki})'$.

Second Stage

In the second stage, we regress the dependent variable Y_i on the predicted values of all endogenous regressors, all exogenous variables and an intercept using OLS.

$$Y_i = \beta_0 + \beta_1 \hat{X}_{1i} + ... + \beta_k \hat{X}_{ki} + \boldsymbol{X}'_{2i}\boldsymbol{\beta}_2 + u_i$$

This gives ($\hat{\beta}_0^{2SLS}$,$\hat{\beta}_1^{2SLS}$, ... $\hat{\beta}_k^{2SLS}$)', the 2SLS estimates of the model coefficients.

$\rightarrow$ All variables in the second stage regression are exogenous because we replaced endogenous regressors by predictions based on only exogenous information.

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Under MLR.1–MLR.2,
instrument exogeneity, $E(\boldsymbol{u}|\boldsymbol{Z}) = 0 \Leftrightarrow E(u_i|Z_{1i}, Z_{2i}, ..., Z_{li}) = 0$,
and instrument relevance,
... the 2SLS estimator is consistent for $\boldsymbol{\beta}$.

Example in R

Multiple Regression with endogenous regressor in R

We are interested in the causal effect of the number of kids on maternal labor supply (in hours per week). «»= data("labsup") load dataset on mothers, incl. number of kids and hours worked table(labsup$kids$)$sampleincludeswomenwithtwoormorekidsols$_{l}absup <$
$-lm(hours\ kids + age + agesq + educ + black + hispan, data = labsup)baselinemodel$@

Multiple Regression with endogenous regressor in R

$$\text{«»= coeftest(ols}_{l} absup, vcov. =$$
$$vcovHC(ols_{l} absup, "HC1"))@WhycouldtheOLSestimatorofthecoefficientonkidsbebiased?$$

Multiple Regression with endogenous regressor in R

We want to use *samesex* as IV, an indicator that takes value 1 if the first two kids of the women have the same sex. Do you think this is a suitable IV?

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IV Relevance

«message = F»= Relevance Test linearHypothesis(first$_I$*absup*, *c*("*samesex* = 0"), *white*.*adjust* = "*hc*1")@

2SLS estimation in MLR

```
> library(estimatr)
> # important:  include all IVs + exogenous regressors behind the | !
> tsls_labsub <- iv_robust(hours ~ kids + age + agesq + educ + black +
hispan | samesex + age + agesq + educ + black + hispan, data = labsup,
se_type = "HC1", diagnostics = T)
> summary(tsls_labsub)
```

Coefficients:

term	estimate	std.error	statistic	p.value	conf.low	conf.high	df
kids	-5.120	3.056	-1.675	0.094	-11.110	0.870	31850
age	2.129	0.483	4.410	0.000	1.183	3.075	31850
...							

Diagnostics:

	numdf	dendf	value	p.value
Weak instruments	1	31850	45.763	1.36e-11 ***